



Marking key

Description	Marks
Task question 1	
Correctly Identifies any errors in the questions AND provides the correct answer for all three index laws	3
Correctly Identifies any errors in the questions involving multiplication and division laws AND provides the correct answer for all three index laws	2
Identifies errors in the use of the multiplication laws AND includes correction	1
Subtotal	3
Answer could include, but is not limited to:	
Simplify in index form	
a) $2^4 \times 2^3 = 2^{12} \times \Rightarrow 2^7$	
b) $3^{10} \div 3^5 = 3^2 \times \Rightarrow 3^5$	
c) $(5^4)^2 = 5^6 \times \Rightarrow 5^8$	
d) $6^4 \times 6^2 \times 6^3 = 6^9 \checkmark$	
e) $(7^3)^4 = 7^{12} \checkmark$	

Task question 2	
Explains the zero law using the division law and then expands to demonstrate the cancelling	2
Explains the zero law using the division law BUT fails to expand and cancel	1
Subtotal	2
Total	5
Answer could include, but is not limited to:	
$\frac{3^2}{3^2} = 3^{(2-2)} = 3^0$ $\frac{3^2}{3^2} = \frac{3 \times 3}{3 \times 3} = 1 \text{ [The 3's cancel out]}$	



Task question 3	
Demonstrate an understanding of the index through a complete expansion process	2
Demonstrates an understanding of the multiplication law but does NOT expand fully	1
Subtotal	2
Total	7
Answer could include, but is not limited to:	
$3^4 \times 3^4 \times 3^4 = 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 = 3^1$	

Task question 4	
Correctly uses the index laws AND able to alter the way numbers are represented through the use of indices and factors AND fully simplifies solution	3
Two of the three behaviours mentioned above	2
One of the three above mentioned behaviours	1
Subtotal	3
Total	10
Answer could include, but is not limited to:	
$\frac{3^4 \times 9}{12} = \frac{3^4 \times 3^2}{3 \times 4}$ $= \frac{3^6}{3 \times 4}$ $= \frac{3^5}{4}$ $(4 \times 2^2)^5 = (2^2 \times 2^2)^5$ $= (2^4)^5$ $= 2^{20}$ or $(4 \times 2^2)^5 = 4^5 \times 2^{10}$ $= (4 \times 2^2)^5$ $= 2^{20}$	